Lab 2: Learning When Labels Are Scarce

1. What is this lab about?

In real security work, labelling traffic is slow and expensive, so you usually have labels for only a small part of your data and a large *unlabeled* data. **Semi-supervised learning (SSL)** uses both the few labels and the unlabelled data together. In this lab you pretend you only have a few labels, use SSL to get value from the rest, and see how close you can get to your Lab 1 detector.

You will work at three label budgets:

- **1% labels** — only a tiny slice is labelled.
- **5% and 10% labels** — a bit more, so you can see the trend.

2. What you will be able to do afterwards

- Set up a fair experiment where only a few labels are available.
- Use unlabelled data to improve a detector (semi-supervised learning).
- Compare against the right baselines so you can tell if it really helped.
- Explain when semi-supervised learning helps and when it backfires.

3. Tools and setup

- **Language:** Python 3. Work in a Jupyter notebook or google colab or plain scripts , you can use what you prefer.
- **Core libraries:** pandas (load/clean data), scikit-learn (models, metrics, and semi-supervised tools), matplotlib (plots).
- **Reuse Lab 1:** build on the code you already wrote in Lab 1. A neural network is only needed if you pick Mean Teacher.
- **Install once:** pip install pandas scikit-learn matplotlib (add tensorflow or torch if you use them).

4. The dataset

Use the **same dataset and the same test set as Lab 1** (CICIDS2017 or CICIDS2018, from the Canadian Institute for Cybersecurity). Reuse your Lab 1 cleaning so the two labs compare fairly.

Make life easier
Work on a sample first, then scale up if you have time. With only 1% labels, the exact rows you pick matter, so if you can, run each setting a few times and report the average.

Reuse your Lab 1 setup (and why)
Keep the same cleaning, the same train / validation / test split, and the same test set.

The test set stays fully labelled and untouched. This is what lets you compare Lab 2 against Lab 1 honestly.

5. Which methods to use

Use **pseudo-labelling** (required), and **one more** semi-supervised method of your choice.

Method	What it is / beginner tip
Pseudo-labelling (required)	Train on the few labels, let the model guess the rest, keep only the confident guesses, and retrain. Simple and effective.
Co-training (choose one)	Split the features into two halves, train two models, and let each one teach the other with its confident guesses. Easiest of the three.
Mean Teacher (choose one)	A student model learns to agree with a slow, averaged teacher, even on slightly noised input. Needs a neural network.
FixMatch (choose one)	Keep a guess only when the model is confident on a lightly changed input, then make it agree on a strongly changed one. Most advanced, and changing table data is tricky, so explain your choice.

Baselines cheat-sheet

Always compare your SSL result against two lines: a model trained on the few labels only (the lower line SSL must beat), and your full-label Lab 1 model (the upper line SSL tries to approach). Without both, an SSL score means nothing.

6. Step-by-step tasks

Step 1: Recreate your baselines

- Note your full Lab 1 score (all labels). This is the upper line. Then, at each budget (1%, 5%, 10%), train a plain model on only the labelled slice. This is the lower line.

Step 2: Make the "few labels" splits

- From the Lab 1 training set, keep the labels for 1%, 5%, and 10% of the rows (stratified, so the class balance stays the same). Hide the labels on the rest, this is your unlabelled pool. Fix the random seed.

Step 3: Pseudo-labelling

- Train on the labelled slice, guess the unlabelled pool, keep only the confident guesses, add them, and retrain. Repeat. Say what confidence cut-off you used.

Step 4: One more method (your choice)

- Add co-training, Mean Teacher, or FixMatch at each budget. Keep everything else the same as Step 3 so the comparison is fair.

Step 5: Run one "change one thing" experiment

What an ablation is

Change exactly one thing, keep everything else the same, and see what happens to the scores. It shows which choices actually matter.

- Pick one: (a) try a few confidence cut-offs
- (b) use 25% / 50% / 100% of the unlabelled pool. Report the before/after numbers.

Step 6: Compare your results:

- Put all results in one table (accuracy, macro-F1, recall, ROC-AUC, FAR), and draw a curve of a metric against the label budget with the lower and upper lines shown.

- **Right metric:** because the data is imbalanced, judge with macro-F1 and FAR, not accuracy alone. Also check recall so you do not miss attacks.

7. What to hand in

Your code. A notebook or scripts plus a short README or PDF that lists the libraries and how to run it. Include the random seed.

A short report (2–3 pages). Suggested sections:

- Why labels are scarce, and what semi-supervised learning is , a short paragraph.
- What you did .
- Results (one clear table + a curve, each with a caption you actually refer to).
- Discussion (did SSL help, at which budget, and where it failed).
- Who did what (one or two lines).

How your work is graded

Every lab is graded 3 / 4 / 5 / F on the same four things. You do not need to be perfect to pass , you need the work to run, to make sense, and to be clearly reported. The table shows what "good" looks like for this lab.

Criterion	Weight	What good work looks like here
Implementation and code quality	30%	Code runs start-to-finish and reproduces your numbers; reuses your Lab 1 code; a README explains how to run it; seed is fixed; reasonably tidy.
Experimental design and methodology	25%	Correct "few labels" split; both baselines shown (few-labels-only and full-label Lab 1); one ablation; the test set is never peeked at.
Results and analysis	20%	Accuracy, macro-F1, recall, ROC-AUC and FAR reported; the gain over the lower line is clear; failure cases discussed.
Report quality	25%	Clear 2–3 pages; captioned table/curve used in the text; who-did-what line included + Appendix: your code (screenshots of each part with comment explaining the code)

Submitting your work

- **Where.** Upload to CANVAS by the deadline: your code (a zip or a repository link) and the report as a PDF. Late submissions are not accepted in general.
- **Group work.** Work together for real. Add a short line at the end of the report saying who did what. If one person does everything, that shows up in individual grades.
- **Second chance.** If a lab does not reach grade 3 you may resubmit it once, before the final reporting date in the schedule.
- **Honesty.** You may use libraries, tutorials, and public code, just cite them.

- **Common mistakes to avoid**

- Skipping the few-labels-only baseline, then you cannot show that SSL helped.
- Setting the confidence cut-off too low, so the model learns from its own bad guesses.
- Judging only by accuracy while attacks get missed, always check recall and macro-F1.

Where to read more

- scikit-learn documentation — “Semi-supervised learning” user guide.
- Lee, D.-H. (2013). Pseudo-Label: The simple and efficient semi-supervised learning method for deep neural networks. ICML Workshop on Challenges in Representation Learning.
- Van Engelen, J. E., & Hoos, H. H. (2020). A survey on semi-supervised learning. Machine Learning, 109(2), 373-440.
- CICIDS2017 download from : https://www.unb.ca/cic/datasets/ids-2017.html or Kaggle
- CICIDS2018 download from : https://www.unb.ca/cic/datasets/ids-2018.html or Kaggle